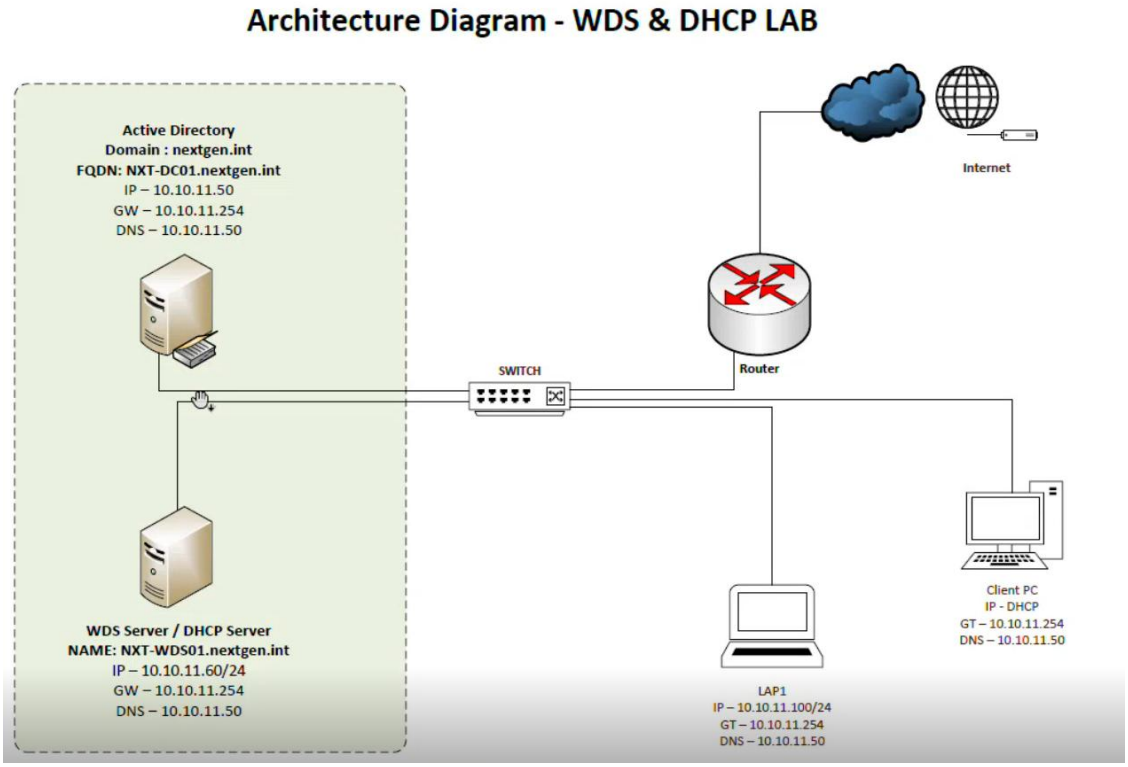
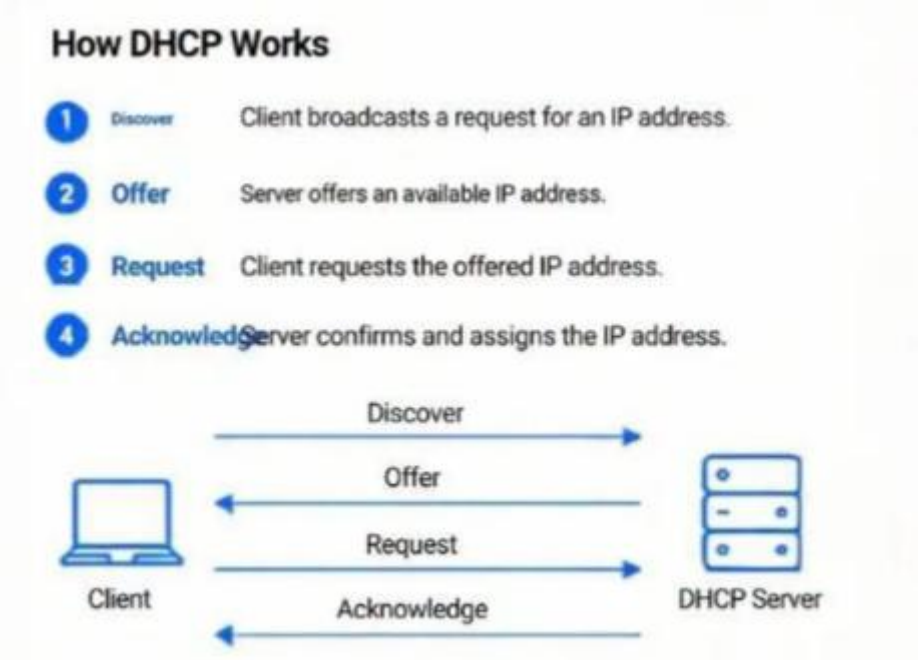
# Part 1 — DHCP (Dynamic Host Configuration Protocol)

## Does PXE Assign an IP Address?

This is a common point of confusion, so it is worth stating clearly: No, PXE does not assign IP addresses. That job belongs entirely to the DHCP server. PXE simply uses the IP address that DHCP hands out in order to talk to the WDS server and download the boot image and installation files.




In our lab, the DHCP server role is installed directly on the Active Directory Domain Controller.




## The DHCP DORA Process

DORA describes the four-step conversation between a client and a DHCP server that results in the client receiving an IP address and its network settings automatically.

| Step | Message | What happens |
|---|---|---|
| **1** | DHCP Discover | The client has no IP address yet, so it broadcasts a Discover message to find any DHCP server on the network. |
| **2** | DHCP Offer | The DHCP server replies with an Offer: an available IP address plus network details such as the subnet mask, default gateway, DNS server, and lease time. |
| **3** | DHCP Request | The client accepts the offered address and sends a Request back to the server, asking to actually use it. |
| **4** | DHCP ACK | The server confirms the lease with an Acknowledge (ACK) message. The client now has an IP address and can communicate on the network. |

**The exchange looks like this between the client and the server:**

```
Client   →   DHCP Discover   →   DHCP Server
Client   ←   DHCP Offer      ←   DHCP Server
```

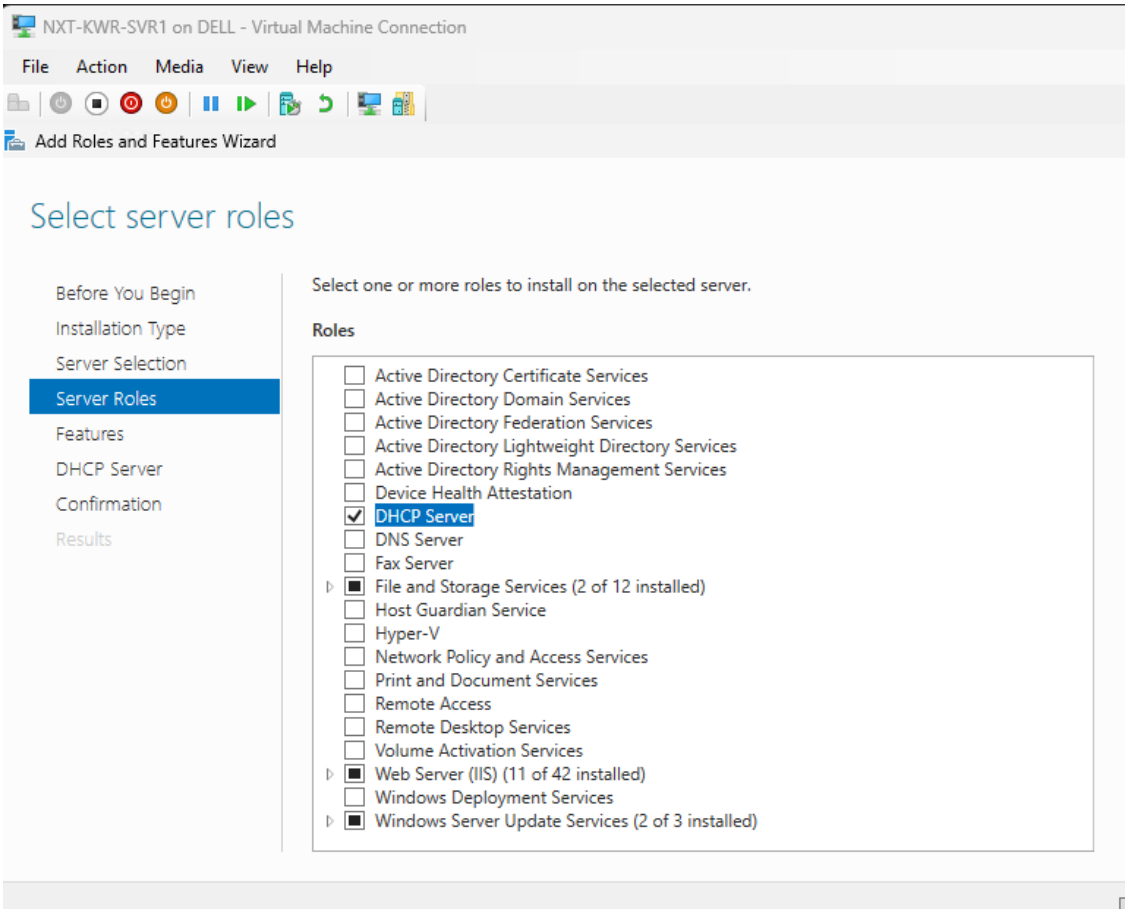
```
Client   →   DHCP Request    →   DHCP Server
Client   ←   DHCP ACK        ←   DHCP Server
```

> *Summary: the DORA process lets a client receive an IP address and every required network setting automatically, with no manual configuration on the client side.*

## Installing the DHCP Server Role

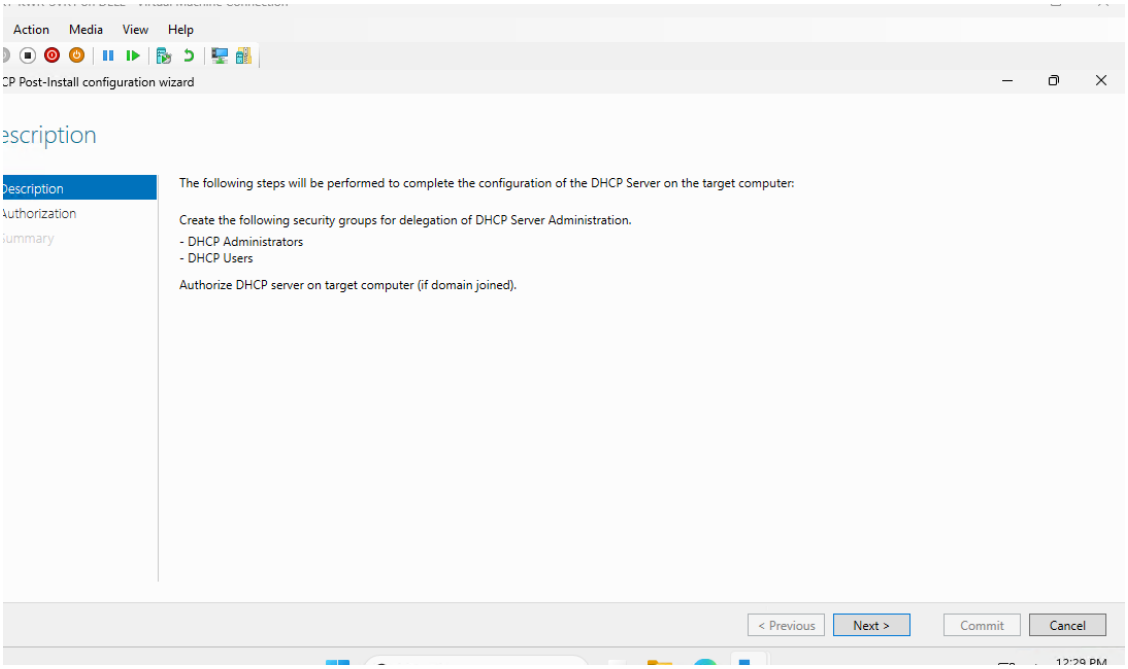
The DHCP server role is installed from Server Manager, inside the Active Directory server in our lab setup.

- Open Server Manager → Add Roles and Features.
- Select the DHCP Server role and continue through the wizard.
- Complete the remaining steps and install. A restart is not required.

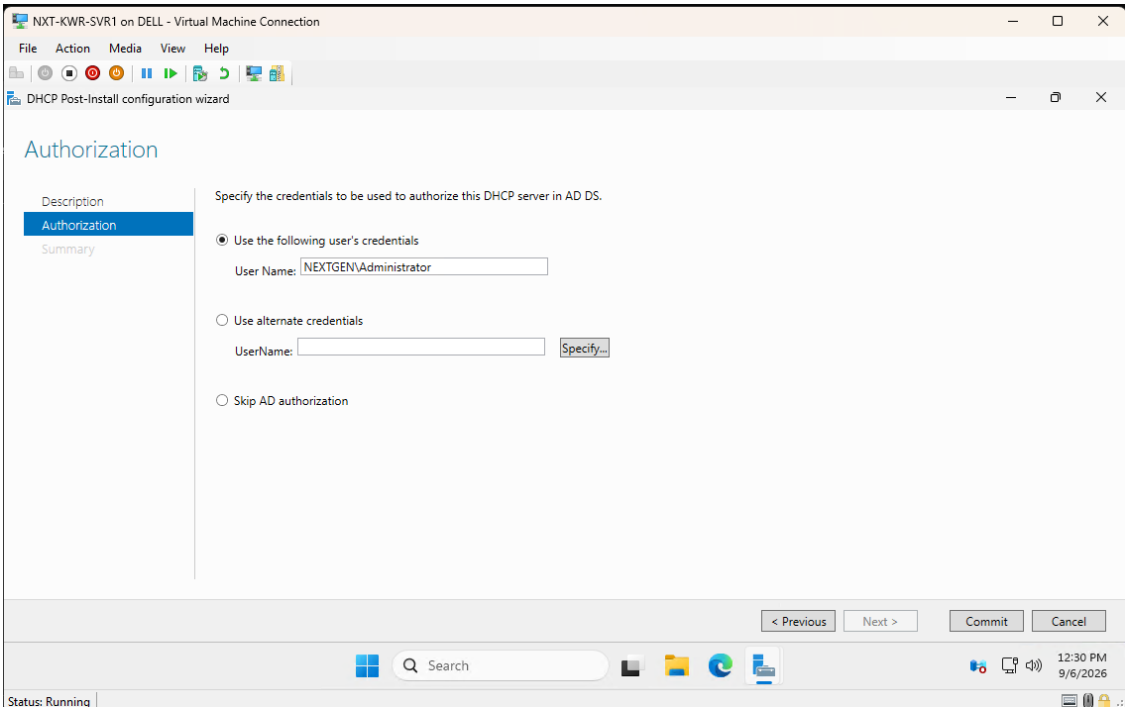



## Post-Install Configuration Wizard

After the role finishes installing, Server Manager prompts you to run the DHCP Post-Install Configuration Wizard. This wizard completes the setup that the role installation alone does not handle: it creates the DHCP Administrators and DHCP Users security groups, and — most importantly — it authorizes the DHCP server inside Active Directory.



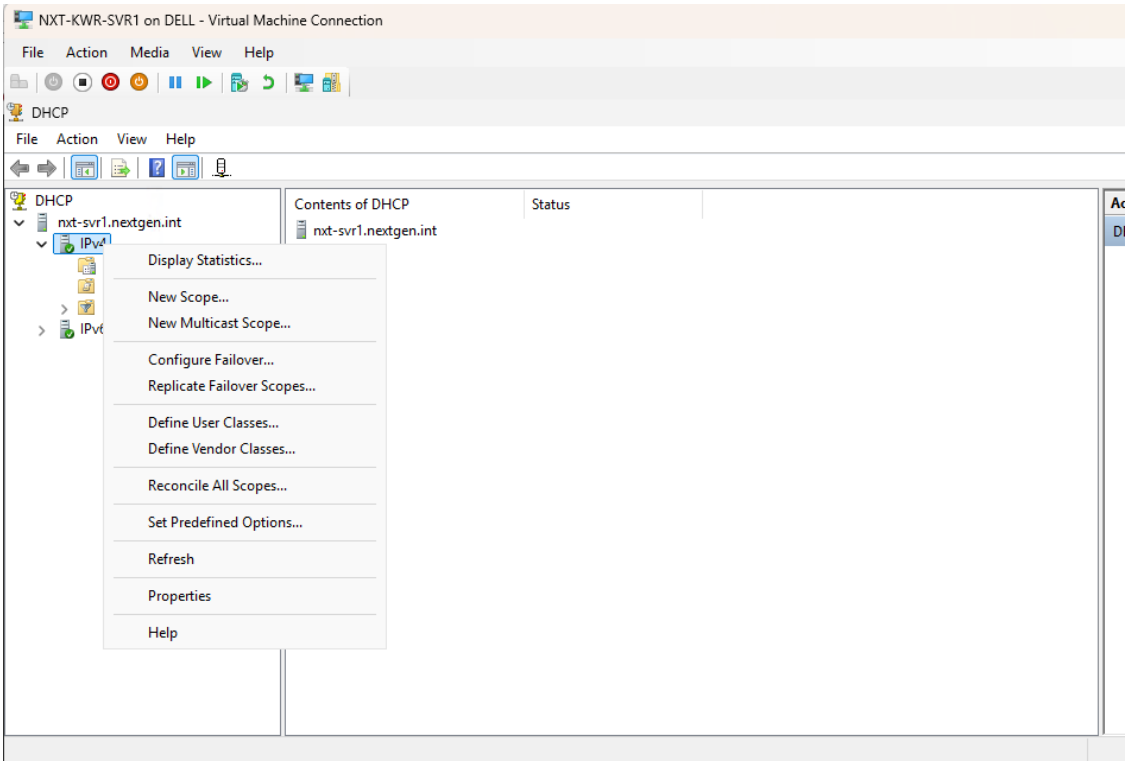

Authorization matters because an unauthorized DHCP server is not permitted to lease IP addresses in a domain environment. This step uses a Domain Administrator account to register the server as a trusted DHCP source in AD. Without it, the server simply will not hand out addresses to domain clients.



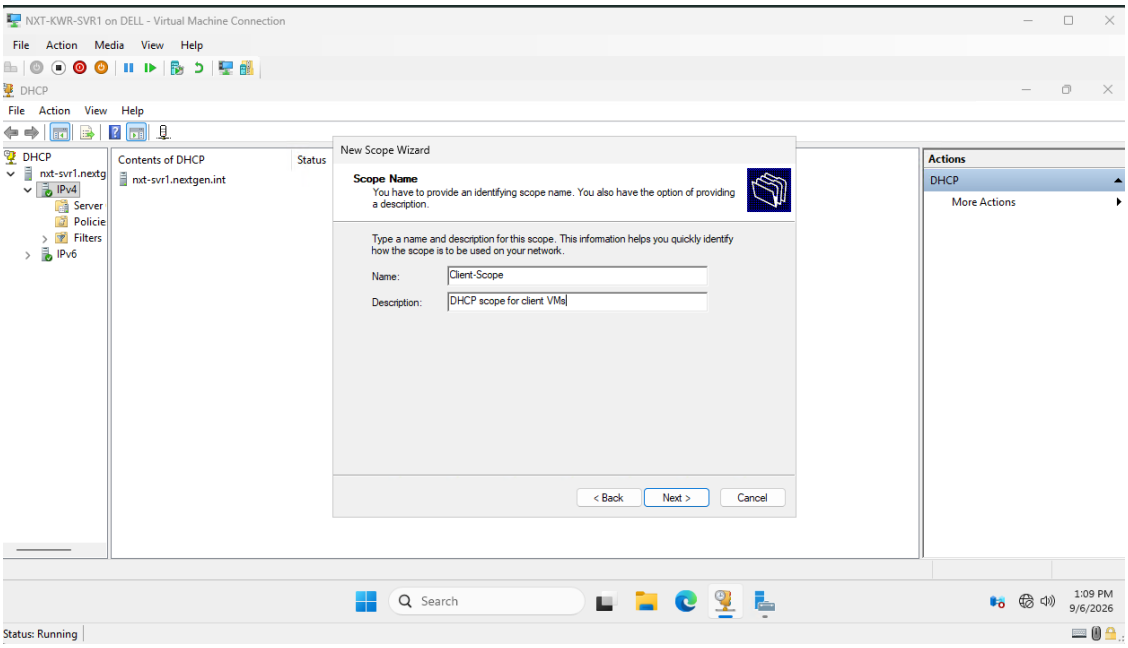

Click Commit, then Close, to finish the wizard.

## Creating a DHCP Scope

With the role installed and authorized, DHCP still cannot hand out any addresses until you create a scope. A scope is a defined range of IP addresses, along with the related settings (subnet mask, default gateway, DNS server, and lease duration), that the DHCP server is allowed to assign to clients. You can open DHCP Manager from Windows Tools in Control Panel, or directly from Server Manager.

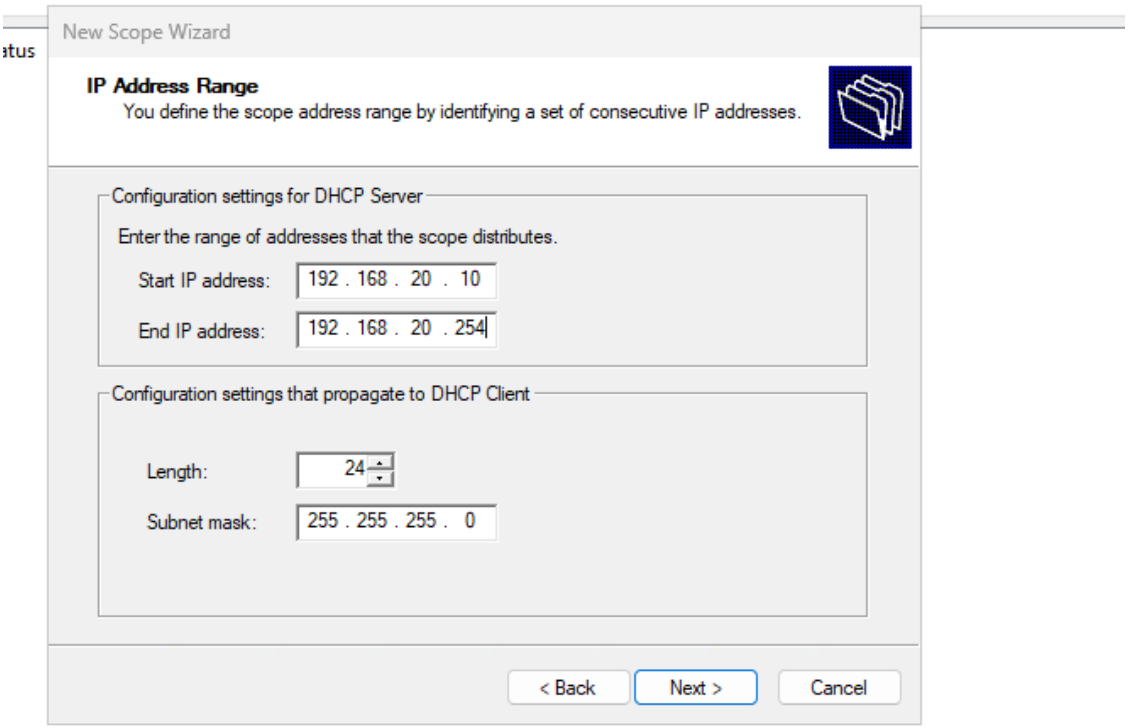



**From DHCP Manager, right-click IPv4 and choose New Scope to launch the scope creation wizard.**



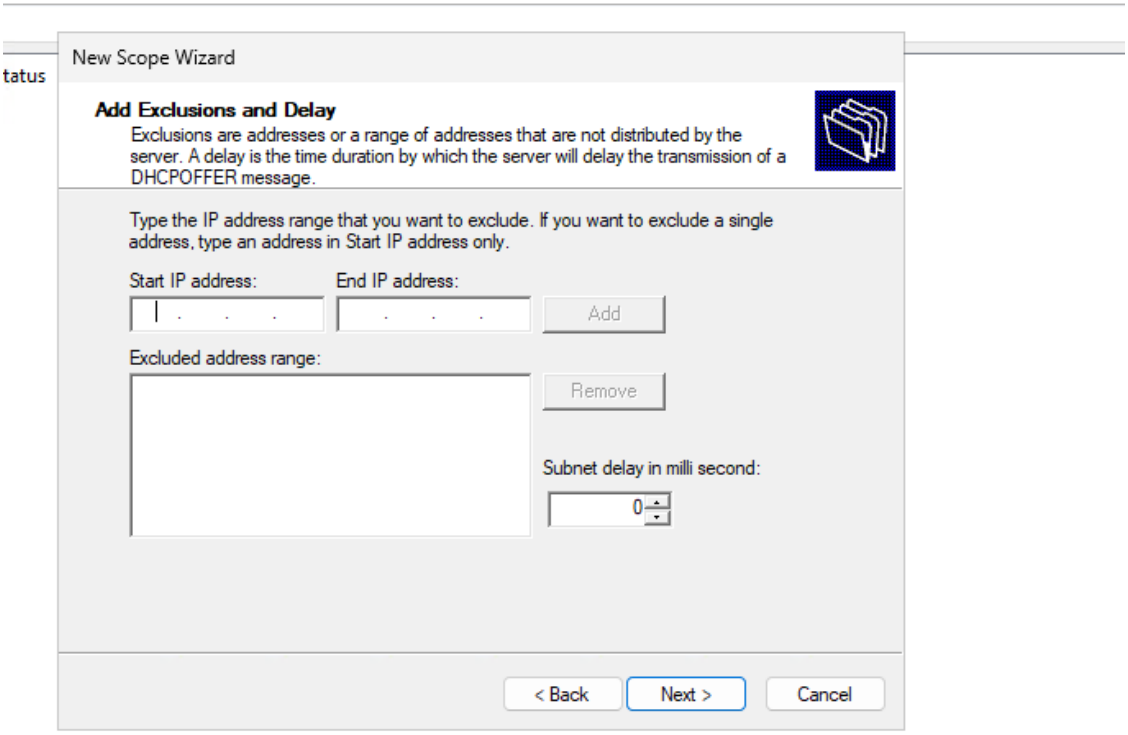
## Step 1: Name and Description

Give the scope a descriptive name and, optionally, a short description — this is just for identifying the scope later, it has no effect on how it functions.



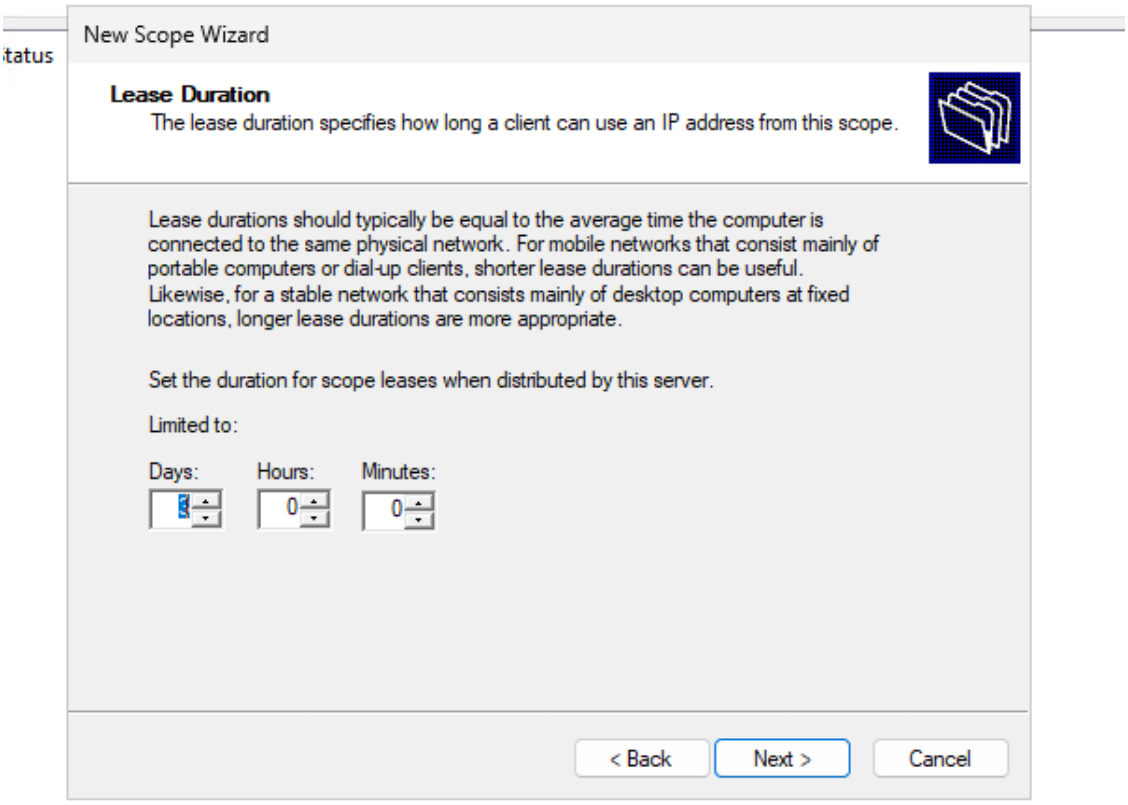

## Step 2: IP Address Range

Enter the starting and ending IP address of the range you want DHCP to manage, along with the subnet mask. The wizard can calculate the subnet mask length automatically once you enter it.



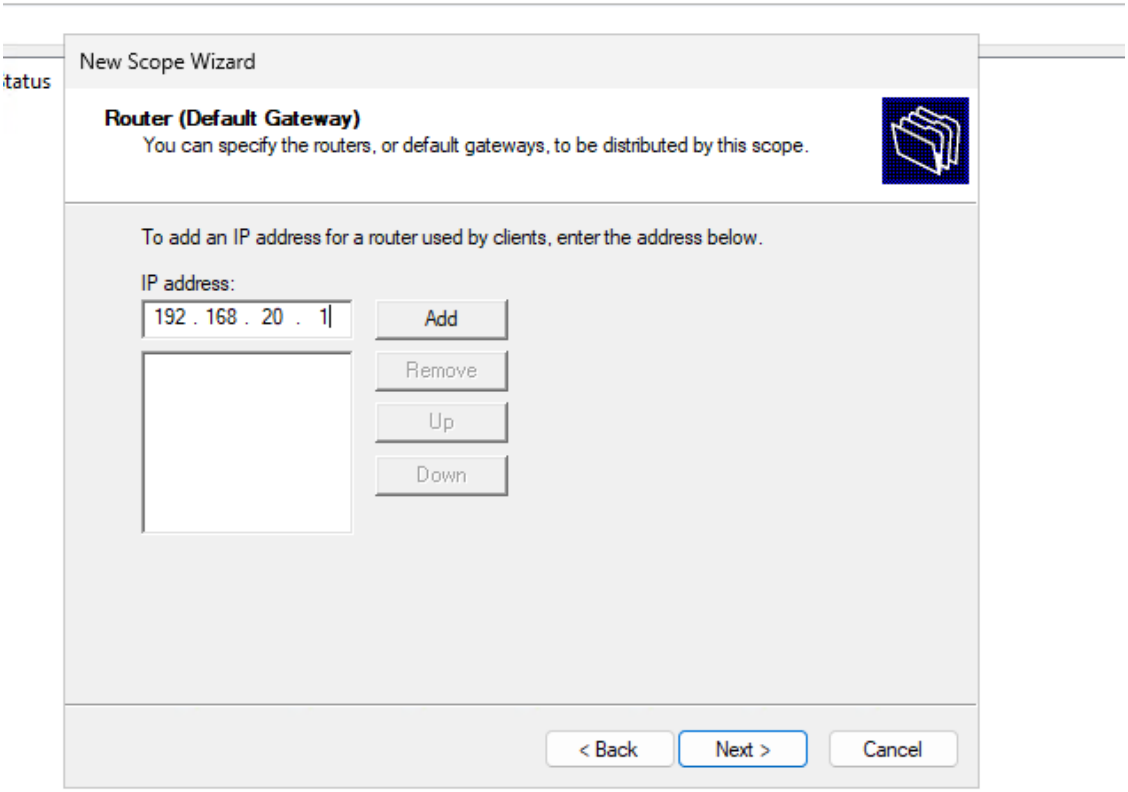

## Step 3: Exclusions and Delay

Exclusions are specific addresses or sub-ranges within the scope that DHCP should never hand out. These are typically reserved for devices that need a fixed, predictable address — servers, routers, and printers, for example. Delay is the amount of time the server waits before sending its DHCP Offer; this is mainly useful when more than one DHCP server exists on the same network and you want to control which one responds first.



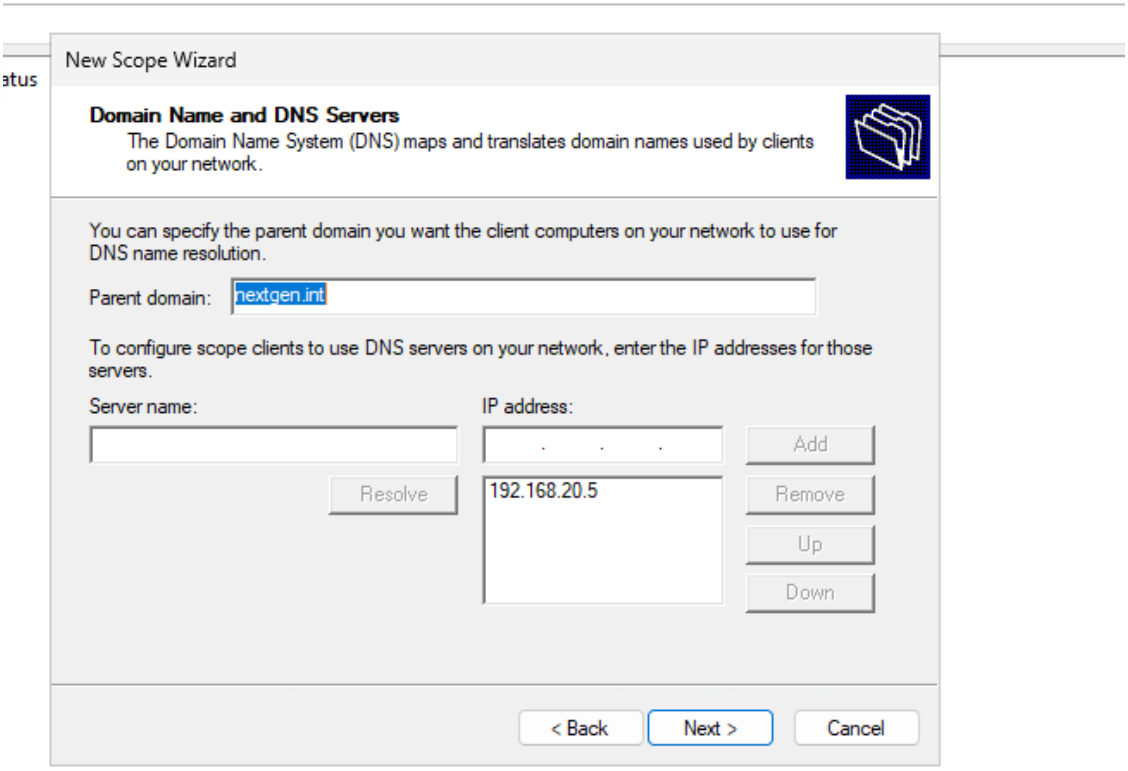

## Step 4: Lease Duration

Lease Duration sets how long a client is allowed to keep an IP address before it must renew it with the DHCP server. Once the lease expires without renewal, the address becomes available for reassignment.




## Step 5: Router (Default Gateway)

Enter the router's IP address here. This is handed to clients as their default gateway, allowing them to reach devices outside their own subnet.



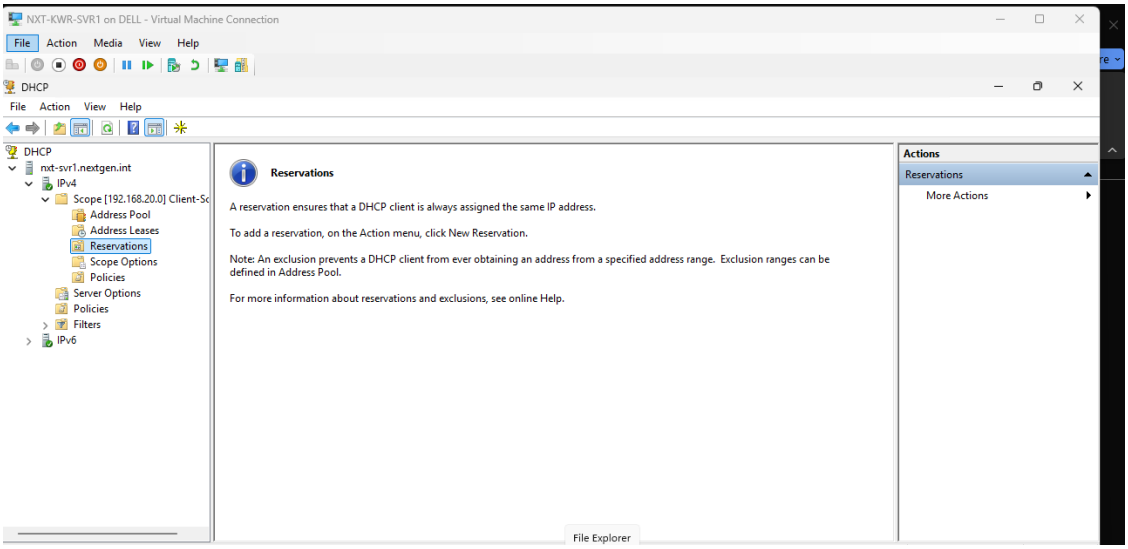

## Step 6: DNS Settings

This step tells clients which DNS server to use and what DNS suffix (parent domain) to apply. In this lab, the domain name is nextgen.int and the DNS server address is 172.20.10.3 — the same machine acting as our Active Directory Domain Controller and DNS server.

The following screen, for WINS servers, is not used in this project, so it can be skipped by clicking Next. On the final screen, choose "Yes, I want to activate this scope now" and click Finish — the scope will not hand out any addresses until it is activated.

## DHCP Reservations

A reservation ties a specific IP address to a specific device's MAC address. The device still goes through the normal DHCP process, but it is guaranteed to receive the same address every time. This is useful for devices where you want DHCP's convenience but also need a consistent, predictable address — printers, IP phones, and other fixed network devices are the typical examples.




## Verifying That DHCP Works

To confirm the scope is working, go to another VM (server or client). If it currently has a static IPv4 address, remove it first. Then disable and re-enable the Ethernet adapter to force it to request a fresh address. Right-click the adapter, choose Status, then Details, and you should see an address from your DHCP range listed there.

